EC3305: SIGNALS AND SYSTEMS

LABORATORY 01

NAME :

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Module Code	
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1. OBSERVATION

1.1 Part 01: Observe the waveform in time domain and in frequency domain

1.1.1 Observations of the sinusoidal waveform

Table 1.1 : Properties of sinusoidal waveform in time domain

Property	Value
Amplitude	
Frequency	
Period	

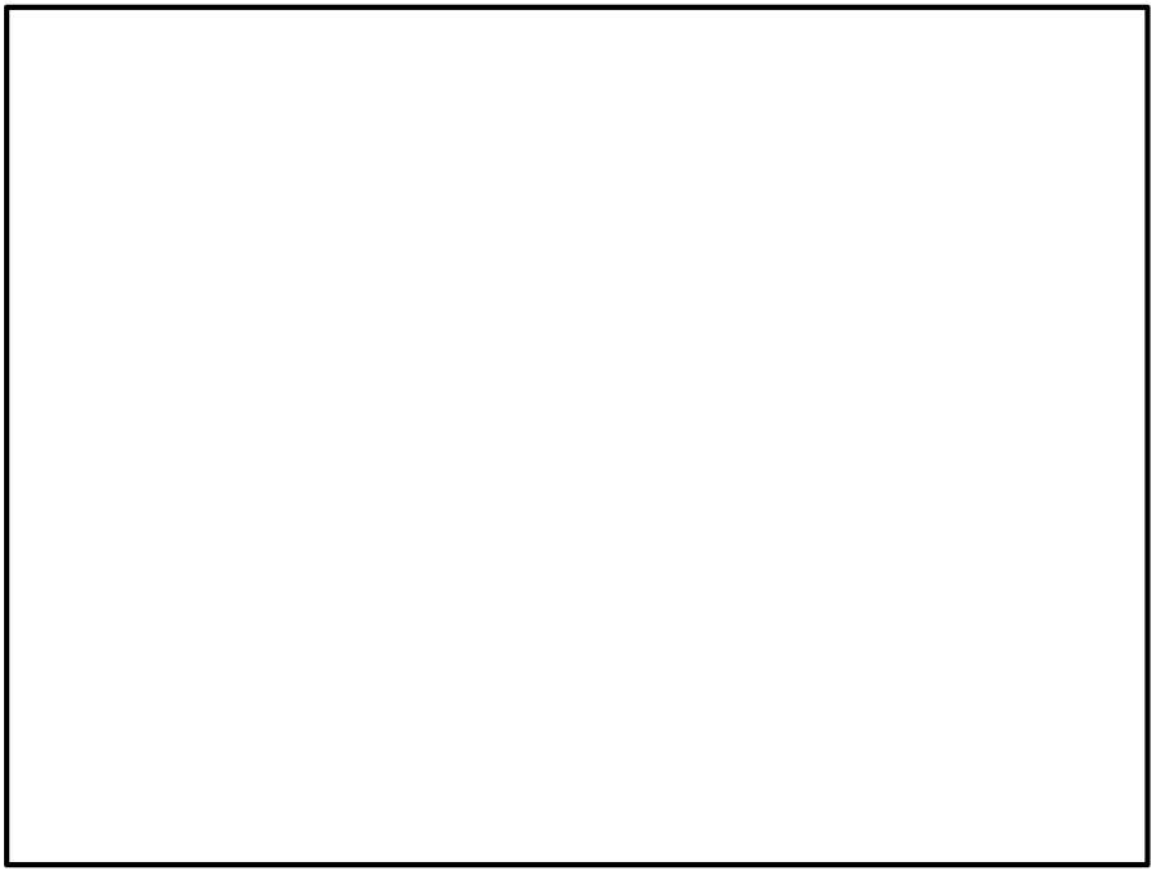


Figure 1.1 : Sine waveform in time domain

Table 1.2 : Properties of sinusoidal waveform in frequency domain

Property	**Value**
Frequency Cener	
Frequency Start	
Frequency Stop	

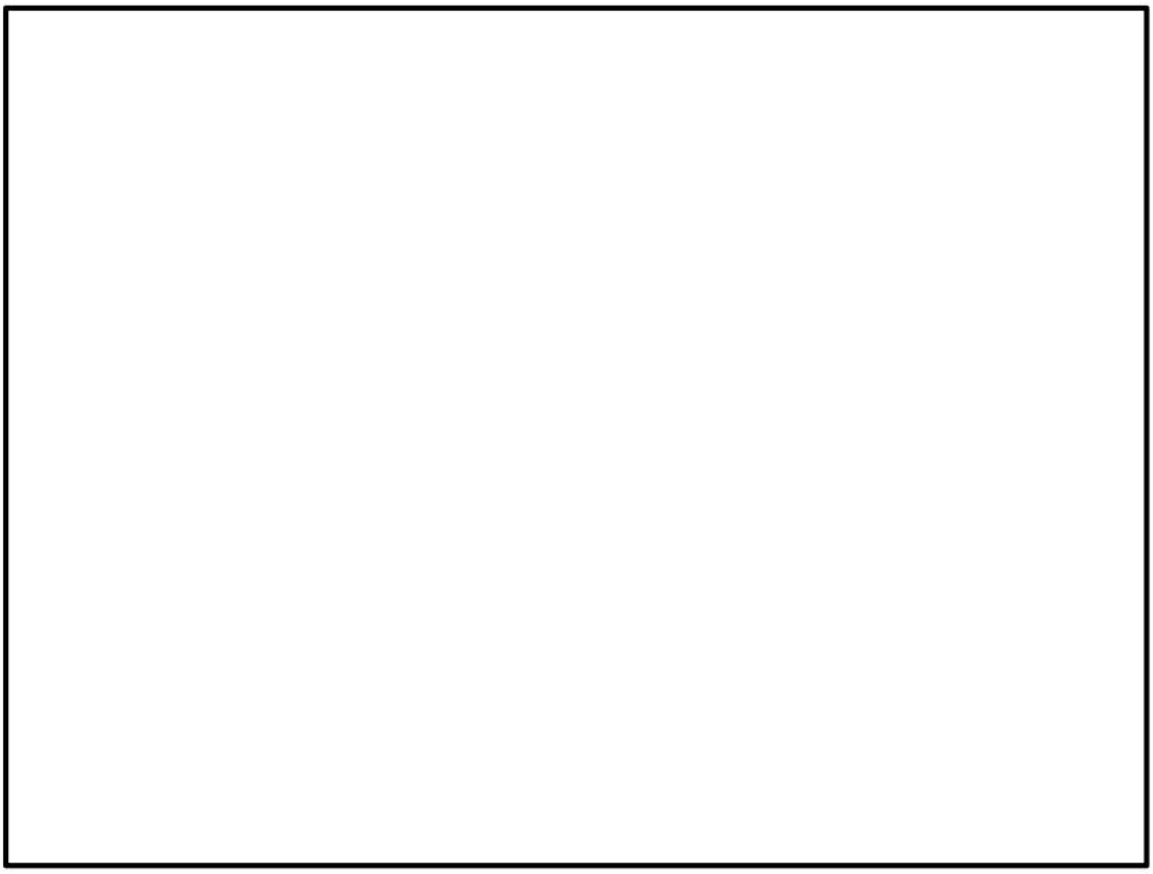


Figure 1.2 : Sine wave in the frequency domain observed by spectrum analyzer

1.1.2 Output of the Adder Circuit

Table 1.3: Properties of Input Waveforms of the Adder Circuit

Signal	Property	Value
1	Peak-Peak Voltage	
	Frequency	
2	Peak-Peak Voltage	
	Frequency	

Table 1.4: Properties of the output waveform of the adder circuit

Property	Value
Peak-Peak Voltage	
Frequency	

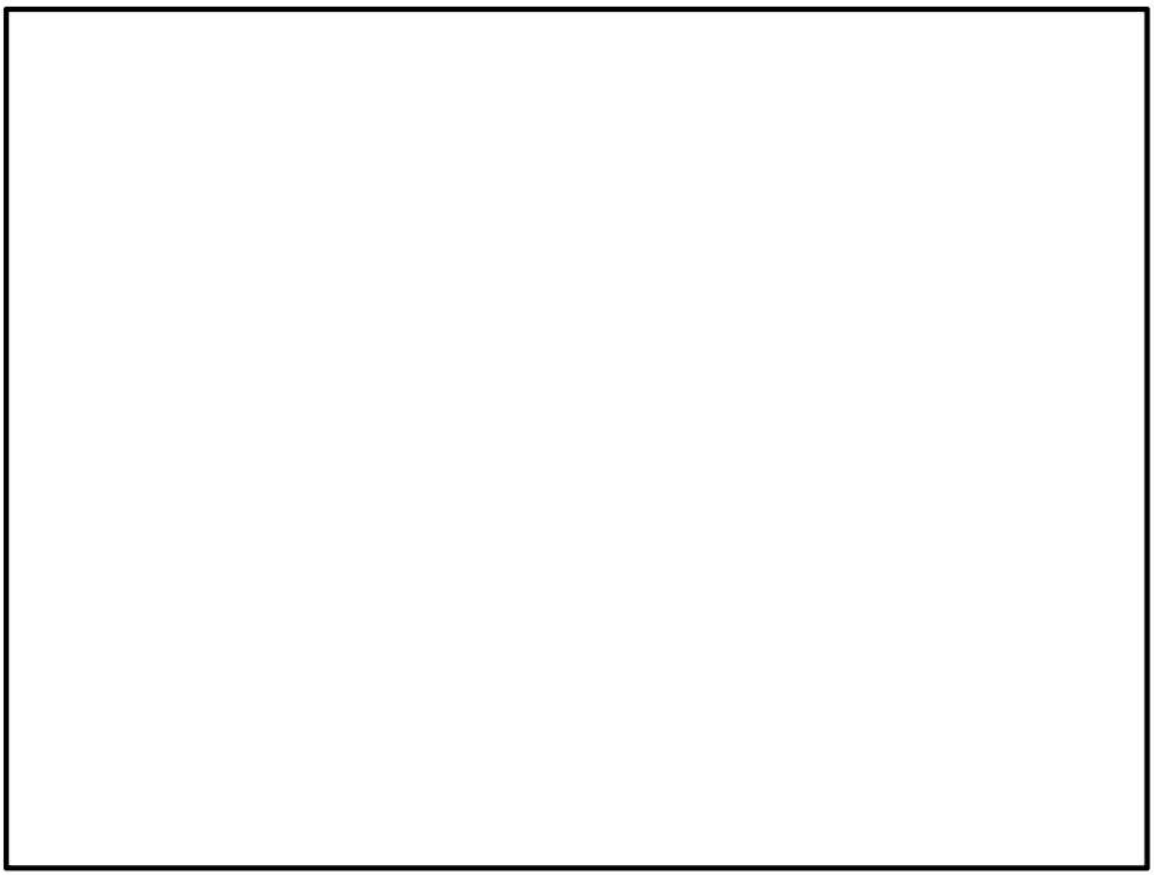


Figure 1.3: Input and output waveform of the adder circuit

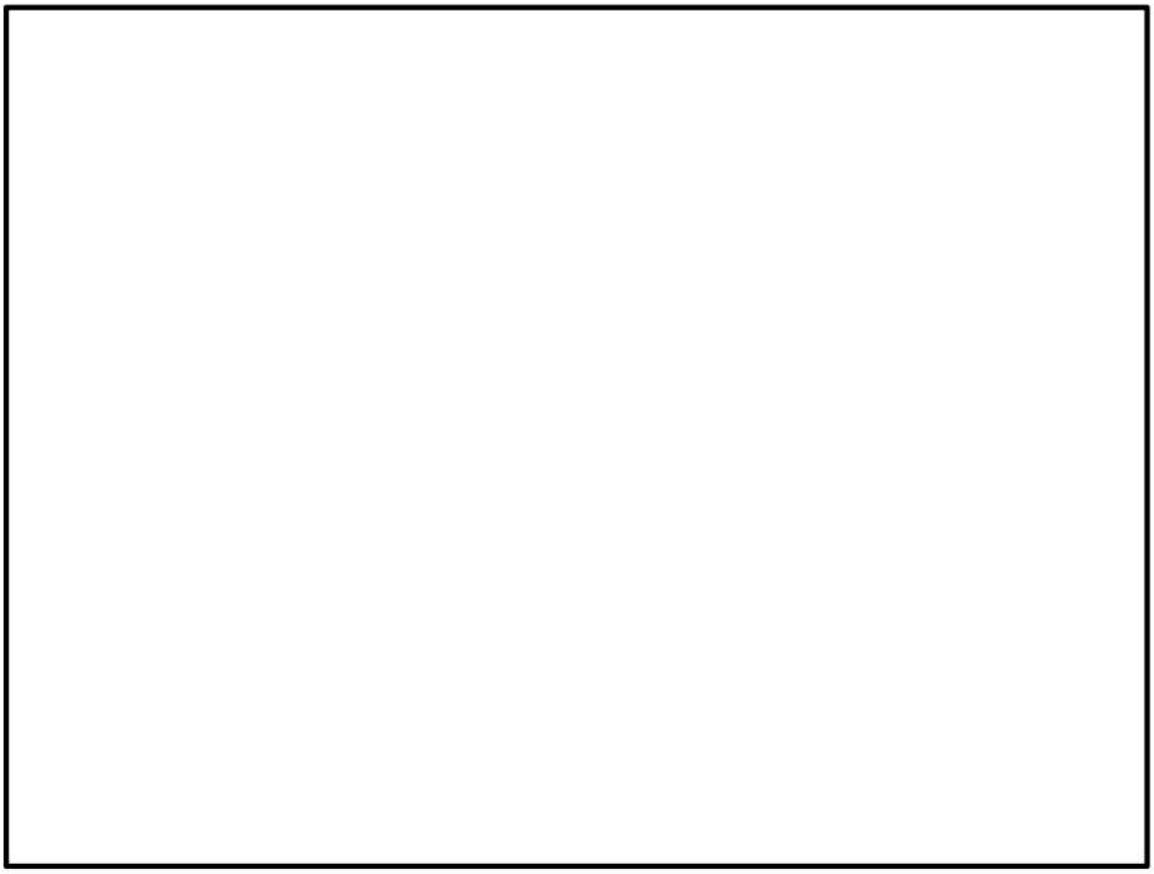


Figure 1.4: Output waveform of the adder circuit in frequency domain

1.1.3 Observations of the square waveform

Table 1.5 : Properties of the square waveform in the time domain

Property	**Value**
Amplitude	
Frequency	
Period	

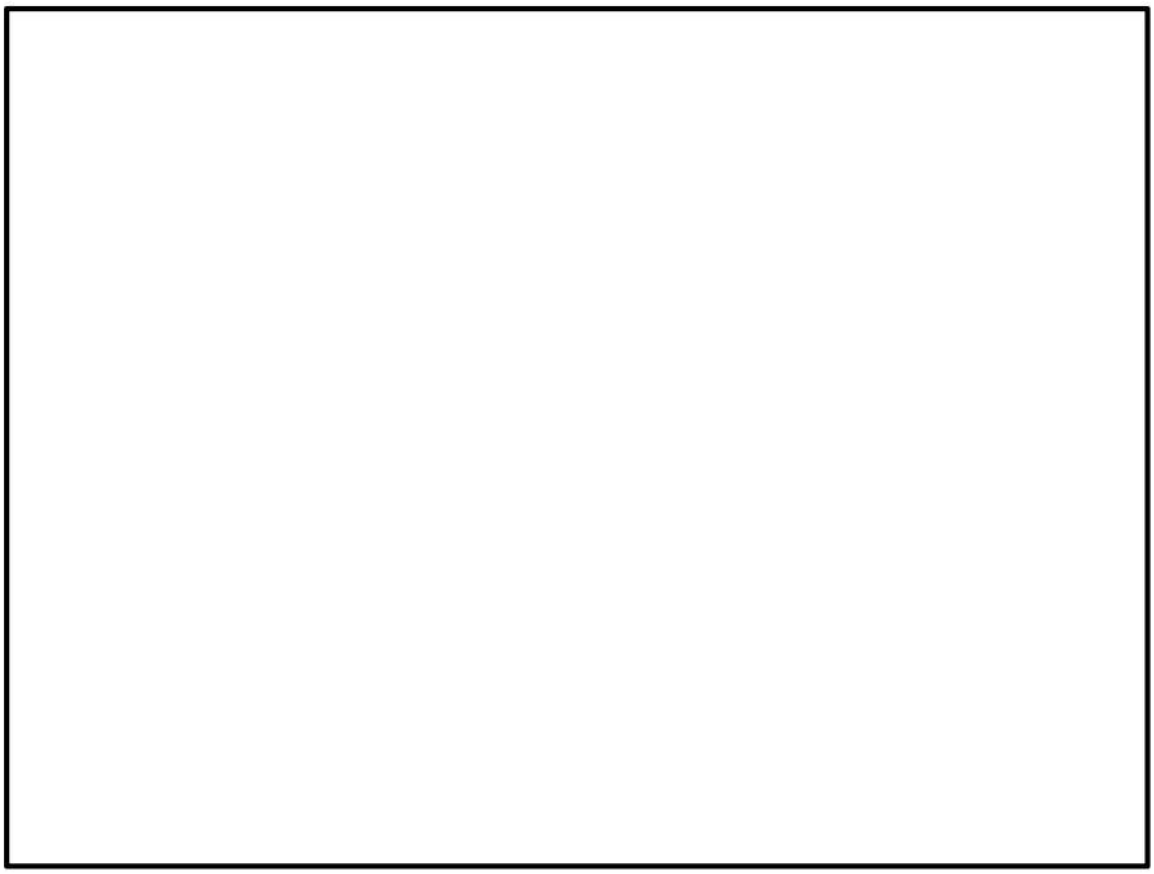


Figure 1.5 : Square wave of duty cycle of 50% in time domain

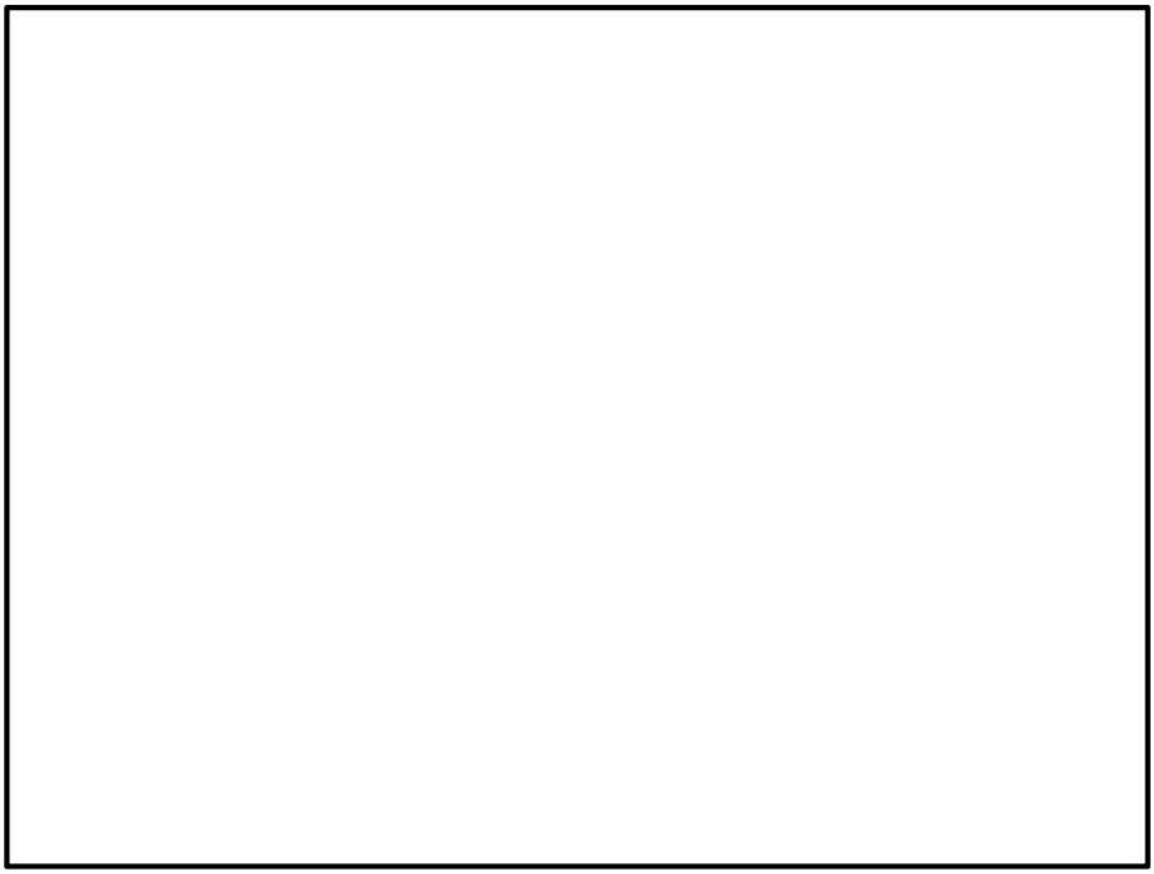


Figure 1.6 : Square wave of duty cycle of 70% in time domain

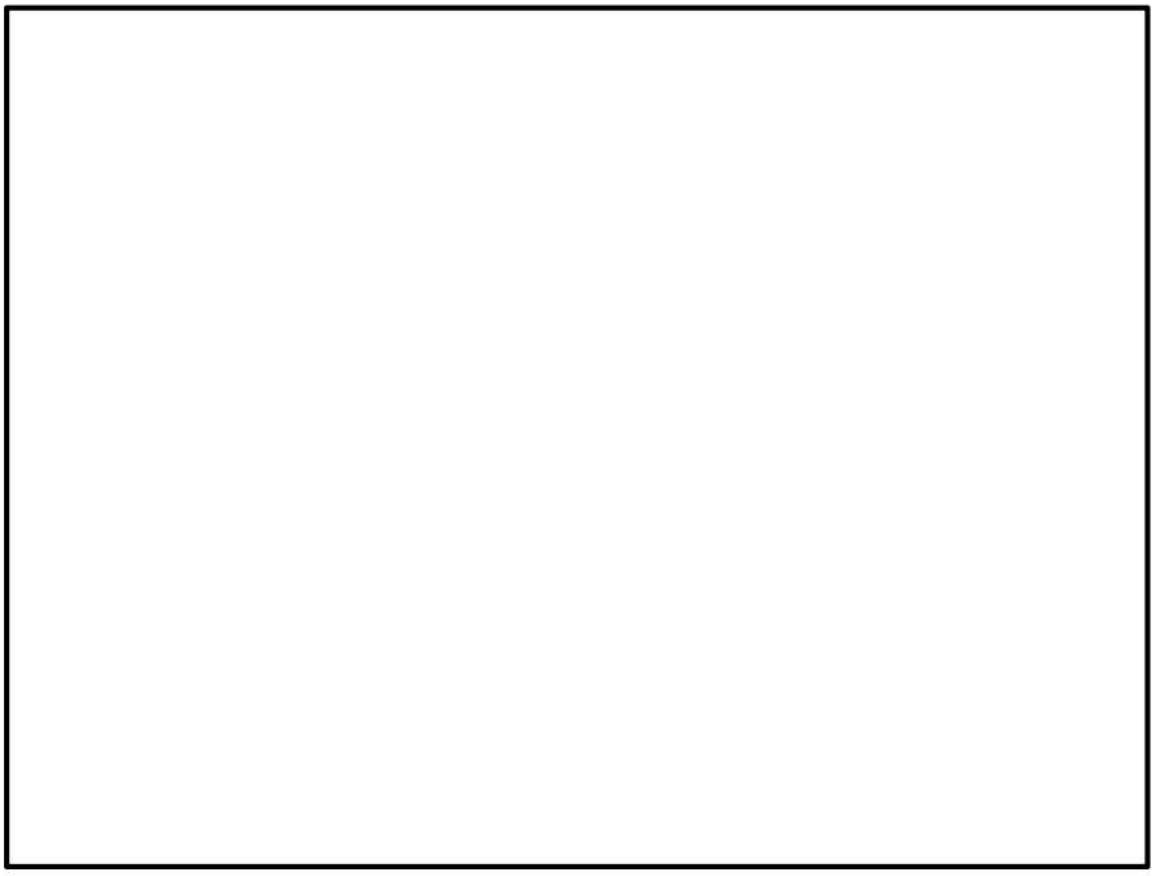


Figure 1.7 : Square wave in the frequency domain observed by spectrum analyzer

1.2 Part 02: Observe the Lissajous pattern using oscilloscope while comparing frequency and the phase shift difference of the two waveforms

Table 1.6 : Properties of the waveforms

Property	Value
Amplitude	
f_x	
f_y	

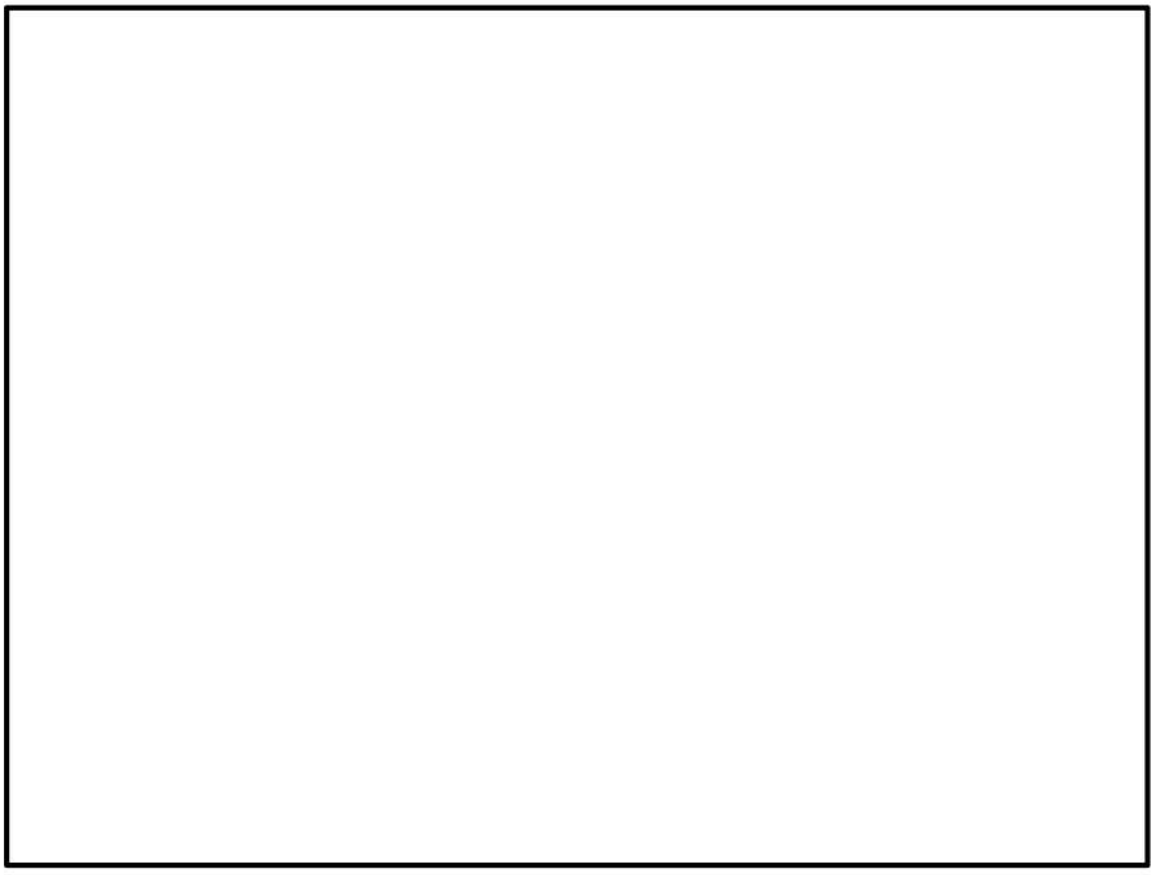


Figure 1.8 : Lissajous pattern for two waves with frequency ratio 2:3 in phase 0°

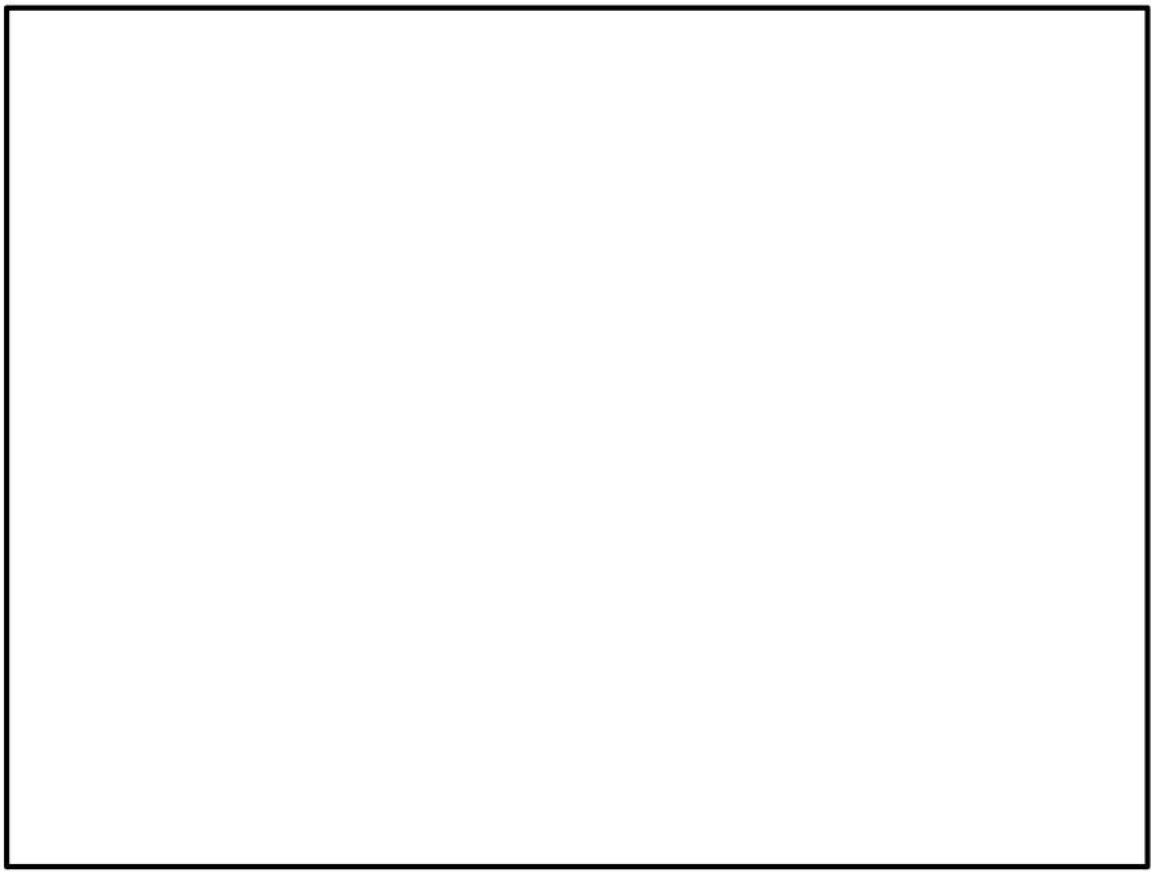


Figure 1.9 : Lissajous pattern for two waves with frequency ratio 2:3 in phase 45°

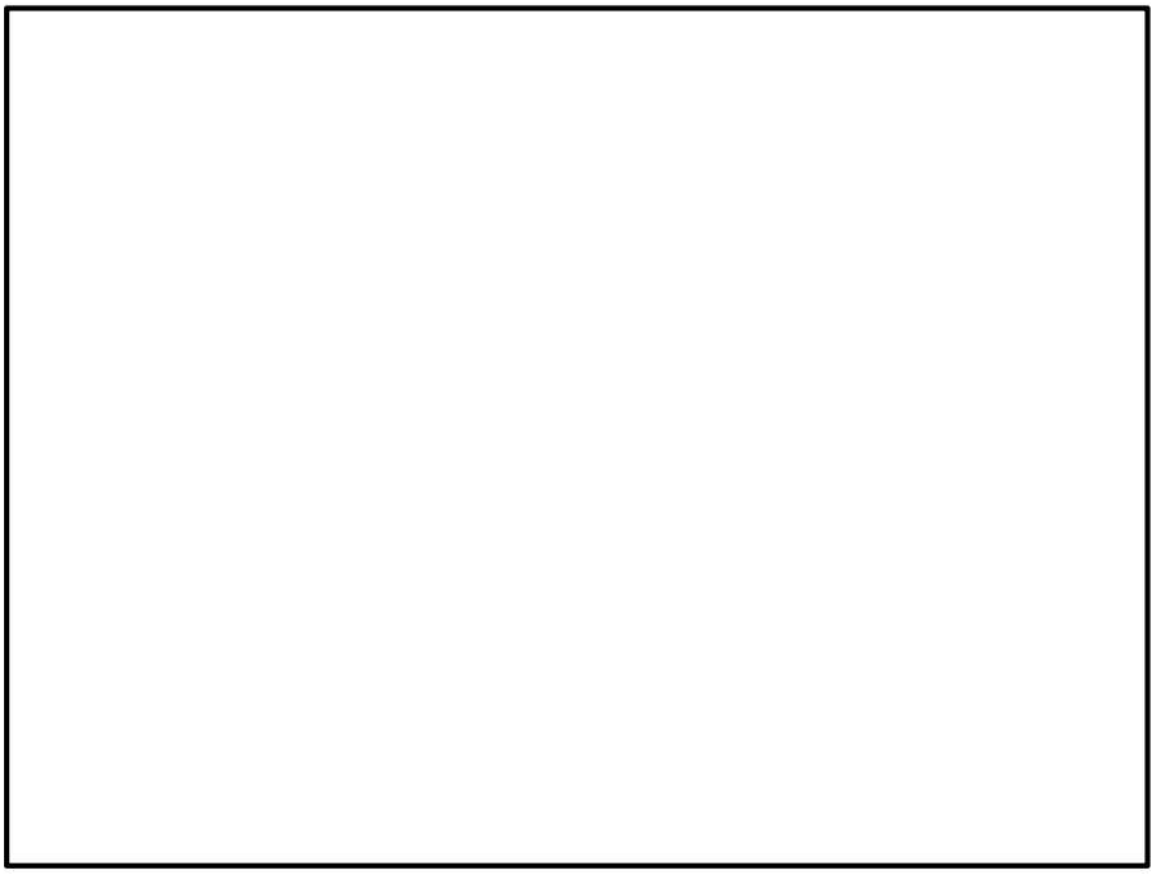


Figure 1.10 : Lissajous pattern for two waves with frequency ratio 2:3 in phase 90°

1.3 Comparisons of Two Waveforms Using Oscilloscope

Table 1.7: Properties of the Input Sawtooth Waveform

Property	**Value**
Amplitude	
Frequency	

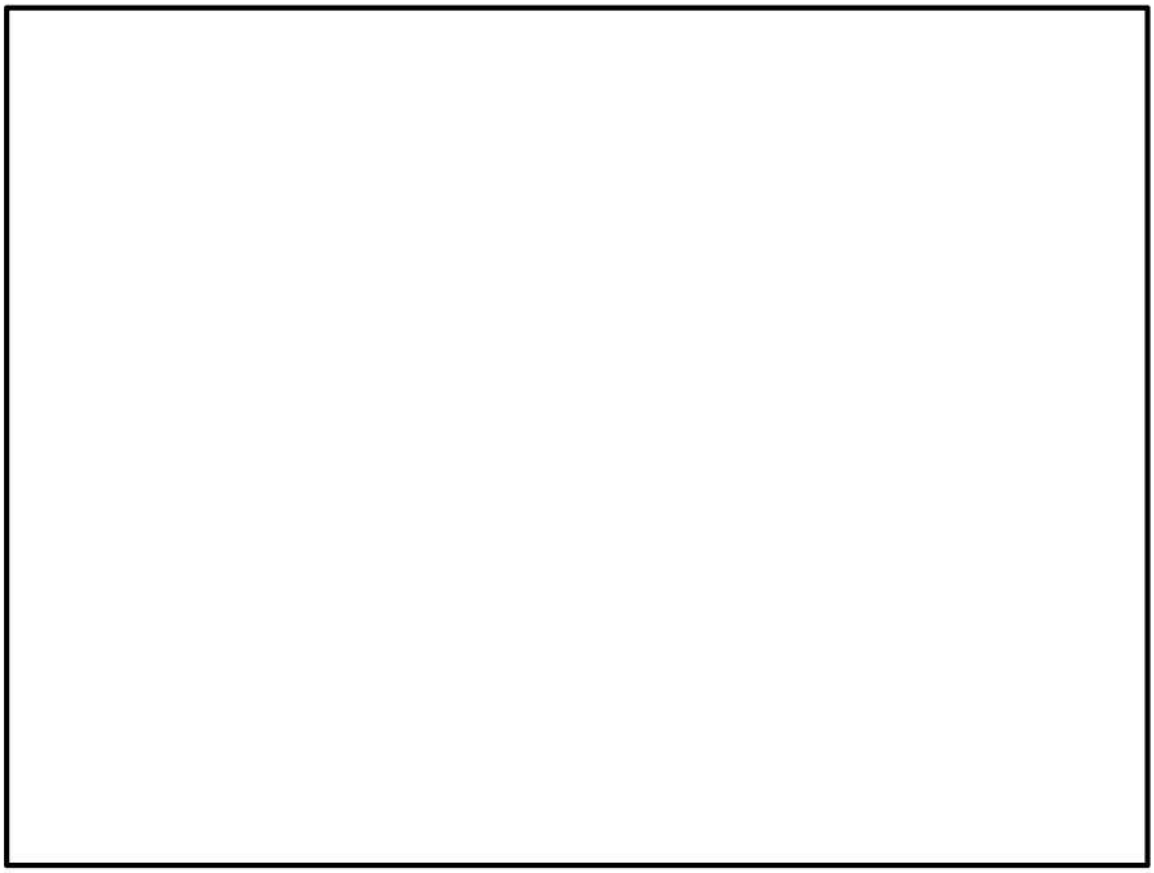


Figure 1.11: Output for generating a square wave through a saw tooth waveform

2. CALCULATIONS

Determine the frequency ratio of the two sinusoidal waveforms with Lissajous pattern.

$$\text{frequency ratio} = \frac{f_y}{f_x}$$

$$= \frac{\text{Number of horizontel crossing}}{\text{number of vertical crossing}}$$

3. DISCUSSION

4. REFERENCES